

Visual and text prompt learning for multi-modal brain disease diagnosis

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Appendix A Method

Appendix A.1 Conv-Transformer3d network

Existing works [1,2] use 2D convolution to extract features by slicing the 3D images. However, these methods ignore the relationship among different slices, which is significant for 3D brain images. Some works [3,4] adopt 3D convolution to capture the 3D structural information, but they do not consider the long-range relationship of slices due to the limits of the convolution operation. To overcome these problems, we propose a memory-efficient backbone, named Conv-Transformer3d. The network is shown in Figure A1. Convolution layer [5] focuses more on the local information within images. Transformer layer [6] attends more to the global relationship among different feature tokens. Based on the characteristics of neuroimaging data from the brain, the low-level details tend to be coarse and noisy, and a spatial smoothing operation is required in preprocessing. Inspired by this, we propose a Conv-Transformer3d as shown in Figure A1. We first introduce a 3D convolution block that includes 3D convolution (3D Conv), layer normalization (LN), rectified linear unit activation function (Relu), and 3D max pooling operation (3DMaxPool). The convolution kernel and stride are set to $3 \times 3 \times 3$ and $1 \times 1 \times 1$ respectively. The stride of 3D max pooling is set to $2 \times 2 \times 2$. We employ two 3D convolution blocks to extract low-level features and reduce the dimensionality of feature maps. This step not only enhances the robustness of feature representations by suppressing noise but also reduces the computational cost. Next, we utilize linear projection, positional encoding, and four transformer blocks [6] to capture the long-range relationship among all feature tokens from different slices, enabling more comprehensive modeling of spatial relationships within the brain. The patch size for linear projection is set to 14×14 . Finally, a mean operation and a fully connected layer are applied to obtain the global feature representation for each subject. The output of the Conv-Transformer3d network for PET and MRI modalities are denoted as:

$$\mathbf{f}_{global}^{pet} = g(\mathbf{H}^{pet}); \mathbf{f}_{global}^{mri} = g(\mathbf{H}^{mri}), \quad (\text{A1})$$

where $g(\cdot)$ denotes the Conv-Transformer3d network. $\mathbf{H}^{pet}$ and $\mathbf{H}^{mri}$ represent the raw data from PET and MRI modalities.

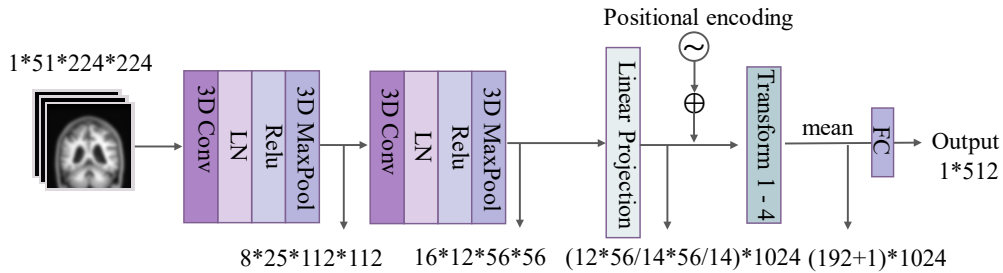


Figure A1 The pipeline of Conv-Transformer3d backbone network to extract modality-specific features.

Appendix A.2 Cross-modality visual prompt module

After freezing the pre-trained PET and MRI models, we propose a novel Cross-modality Visual Prompt Learning module (CVPL) with few additional training parameters to fine-tune the pre-trained PET and MRI models for adapting the multi-modal task. In this module, we design a Consistency-constrained Self-Attention block (CSA) to fully explore the

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complementary information between the pre-trained PET and MRI features. We add CSA at each pre-trained modality transformer block to progressively model stage-wise interaction. To bridge the interactions across different layers, we add a new transformer block after each CSA block.

Consistency-constrained self-attention. We propose a CSA block inspired by self-attention [6]. For the feature of PET or MRI from the i -th transformer block in the pre-trained PET and MRI models, we denote it as $g_i(\mathbf{H}^{pet})$ and $g_i(\mathbf{H}^{mri})$. Firstly, we concatenate the PET and MRI features from the i -th transformer block. Then, we use the self-attention layer to achieve the interaction and fusion between different modalities. As illustrated in Figure A2, we can observe that the self-attention operation enables PET tokens to attend to tokens from MRI and itself, while MRI tokens attend to tokens from PET and itself. After performing a self-attention operation, we split the features at the token level to obtain the interaction features of PET and MRI, denoted as $\mathbf{F}_i^{pet}$ and $\mathbf{F}_i^{mri}$. Assuming that the self-attention operation can perform interaction and fusion between the self-modality and other modality perfectly, the feature $\mathbf{F}_i^{pet}$ will be equal to the feature $\mathbf{F}_i^{mri}$. Based on the assumption, we introduce the Mean Squared Error (MSE) loss function to facilitate the consistency between $\mathbf{F}_i^{pet}$ and $\mathbf{F}_i^{mri}$. The consistency constraint $\mathcal{L}_i^{CSA}$ of the i -th transformer block is defined as:

$$\mathbf{F}_i^{pet}, \mathbf{F}_i^{mri} = \text{split}(\text{Att}(\mathbf{M}_i, \mathbf{M}_i; \mathbf{M}_i)), \text{ where } \mathbf{M}_i = [g_i(\mathbf{H}^{pet}); g_i(\mathbf{H}^{mri})], \quad (\text{A2})$$

$$\mathcal{L}_i^{CSA} = (\|\mathbf{F}_i^{pet} - \mathbf{F}_i^{mri}\|_2^2), \quad (\text{A3})$$

where $[\cdot; \cdot]$ denotes the concatenation operation at the token level. The $\text{Att}(\cdot)$ is defined at Eq. A7. $\text{split}(\cdot)$ represents a split operation. $\|\cdot\|_2$ denotes the Frobenius norm. Finally, the global feature of multi-modal $\mathbf{f}_{global}^{mu}$ can be calculated as:

$$\mathbf{C}_n^{mu} = \text{transformer}(\mathbf{C}_{n-1}^{mu} + \frac{1}{2}(\mathbf{F}_n^{pet} + \mathbf{F}_n^{mri})), \quad (\text{A4})$$

$$\mathbf{f}_{global}^{mu} = FC(LN(\text{mean}(\mathbf{C}_n^{mu}))), \quad (\text{A5})$$

where $\mathbf{C}_n^{mu}$ denotes the output of CVPL. $\text{transformer}(\cdot)$, $FC(\cdot)$, $LN(\cdot)$, and $\text{mean}(\cdot)$ represent the transformer block [6], fully connected layer, layer normalization, and mean operation respectively. n is the number of transformer blocks in the Conv-Transformer3d network.

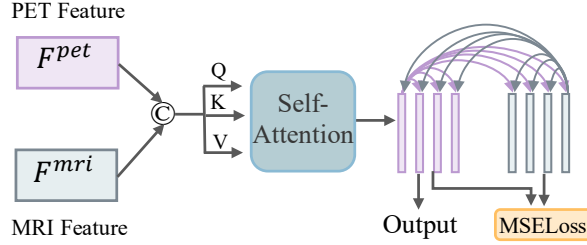


Figure A2 The details of the proposed Consistency-constrained Self-Attention block.

Appendix A.3 Region and Text Prompt Learning

In brain disease diagnosis, only specific neuroimaging regions are strongly associated with the disease. Therefore, it is meaningful to obtain the feature representation of these distinctive regions. Expert knowledge usually provides critical information for brain disease diagnosis. For Alzheimer's disease, reduced glucose metabolism is commonly observed in the temporal and parietal lobes on PET images, while atrophy in the hippocampus and frontal lobes is typically detected on MRI images. Building upon this medical knowledge, we propose RTPL by introducing expert knowledge into the visual learning network to capture the semantic-aware local features representation for brain disease diagnosis. Specifically, we first utilize a CLIP model [7] to encode the hand-crafted expert knowledge prompts. Then, we introduce text and region prompts to capture the task-specific text and visual feature representation. Finally, we employ bidirectional cross-attention to achieve cross-modal mutual guidance and alignment between the text and visual features. The pipeline is shown in Figure A3 (a).

Expert knowledge prompts. To obtain the task-specific text feature representation of expert knowledge, we first hand-craft the expert knowledge prompts from medical professionals, i.e., "Temporal lobe glucose metabolism status; Parietal regions glucose metabolism status; Hippocampus region status; Frontal lobe region status;". We hope the network focuses more on these distinctive regions. In Figure A3 (b), we show these regions of interest through the BrainNet Viewer toolbox [8]. Then, we use the CLIP [7] model to encode the expert knowledge prompts. CLIP is trained on a variety of image-text data, which are collected from natural images. It can predict the relevant text that relates to images. In our work, we only utilize the text-encoder network to obtain the feature representation of expert knowledge, denoted as $\mathbf{X} \in \mathbb{R}^{N \times D}$, where N is the number of expert knowledge prompts and D denotes the feature dimension. Due to the difference between natural language and medical images, we introduce learnable text-prompt vectors $\mathbf{T} \in \mathbb{R}^{N \times D}$ to fine-tune the target task. Then, we can obtain the task-specific text representation $\mathbf{F}^t \in \mathbb{R}^{N \times D}$ of expert knowledge, which is defined as:

$$\mathbf{F}^t = \text{CLIP}(\mathbf{X}) + \alpha \mathbf{T}, \quad (\text{A6})$$

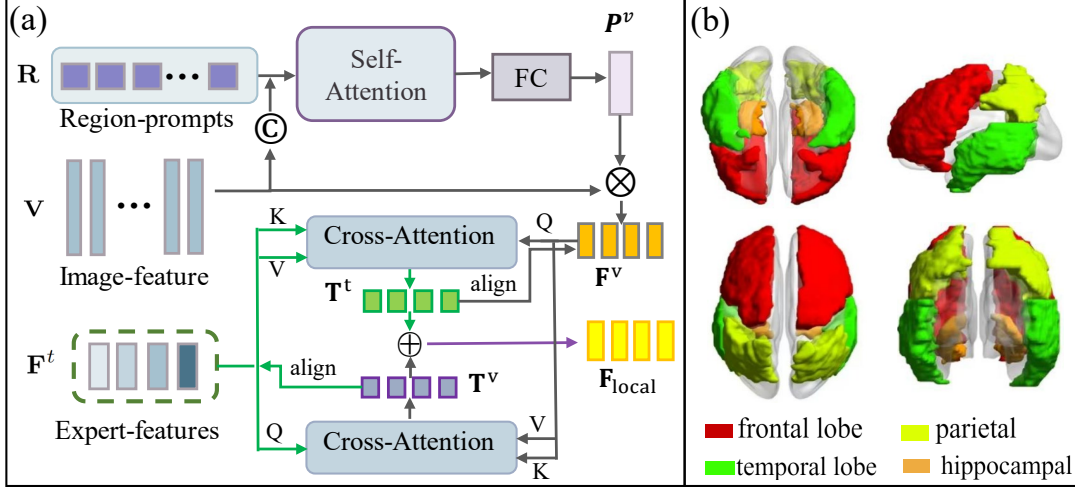


Figure A3 (a) The pipeline of Region and Text Prompt learning module (RTPL). (b) Discriminative regions of interest (ROIs) in the brain based on expert knowledge.

where $CLIP(\cdot)$ denotes the text-encoder network of CLIP, α is a scale factor.

Region-prompt. To capture the semantic-aware local features representation, we introduce region prompts represented as $\mathbf{R} \in \mathbb{R}^{N \times D}$, where N is the number of expert knowledge prompts. For the input of visual features, we denote them as $\mathbf{V} \in \mathbb{R}^{M \times D}$, where M is the number of visual tokens and D is the dimension of the visual feature. We first concatenate the region-prompt vectors and visual features. Then, we adopt the self-attention, fully connected layer to learn the projection matrix. The projection matrix $\mathbf{P}^v \in \mathbb{R}^{N \times M}$ is denoted as:

$$\mathbf{P}^v = FC\left(Att([\mathbf{V}; \mathbf{R}], [\mathbf{V}; \mathbf{R}], [\mathbf{V}; \mathbf{R}])\right);$$

$$\text{where } Att(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = softmax\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D}}\right)\mathbf{V}, \quad (\text{A7})$$

where $[\cdot; \cdot]$ denotes the concatenation operation at the token level. $FC(\cdot)$ denotes fully connected layer. After the projection operation, we can obtain the task-specific visual feature representation $\mathbf{F}^v \in \mathbb{R}^{N \times D}$ as follows:

$$\mathbf{F}^v = \mathbf{P}^v \cdot \mathbf{V}, \quad (\text{A8})$$

where $\mathbf{P}^v$ is defined by Eq. A7, and $\mathbf{V}$ denotes the visual features input into RTPL.

Cross-modal mutual guidance. We utilize bidirectional cross-attention operation to achieve mutual learning and alignment between task-specific text feature $\mathbf{F}^t$ and visual feature $\mathbf{F}^v$. Finally, the semantic-aware local feature with prior expert knowledge $\mathbf{F}_{local}$ can be formulated as:

$$\mathbf{T}^t = Att(\mathbf{F}^v, \mathbf{F}^t, \mathbf{F}^t), \quad \mathbf{T}^v = Att(\mathbf{F}^t, \mathbf{F}^v, \mathbf{F}^v), \quad \mathbf{F}_{local} = \mathbf{T}^t + \mathbf{T}^v, \quad (\text{A9})$$

where $Att(\cdot)$ is defined as Eq. A7. We adopt Visual-to-Text Alignment (VTA) and Text-to-Visual Alignment (TVA) losses to encourage cross-modal guidance and alignment between the visual and text features. The alignment loss $\mathcal{L}_{align}$ can be formulated as:

$$\mathcal{L}_{VTA} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(sim(\mathbf{T}_i^t, \mathbf{F}_i^v)/\tau)}{\sum_{k=1}^N \exp(sim(\mathbf{T}_i^t, \mathbf{F}_k^v)/\tau)},$$

$$\mathcal{L}_{TVA} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(sim(\mathbf{T}_i^v, \mathbf{F}_i^t)/\tau)}{\sum_{k=1}^N \exp(sim(\mathbf{T}_i^v, \mathbf{F}_k^t)/\tau)}, \quad (\text{A10})$$

$$\mathcal{L}_{align}(\mathbf{V}) = \frac{1}{2}(\mathcal{L}_{VTA} + \mathcal{L}_{TVA}),$$

where $sim(\cdot)$ denotes the cosine similarity. τ is the temperature hyperparameter. N is the number of text tokens. For the visual features from PET, MRI and multi-modal branches, we can get the local feature representation through the RTPL module, denoted as $\mathbf{F}_{local}^{pet}$, $\mathbf{F}_{local}^{mri}$ and $\mathbf{F}_{local}^{mu}$.

Appendix A.4 Fusion of global and local features

To fuse the local and global features, we adopt a fusion method at the decision level. For the output from each branch in mVTPrompt, we can obtain the global feature $\mathbf{f}_{global}$ and local feature $\mathbf{F}_{local}$. The prediction of the global and local features is defined as:

$$\mathbf{p}_g = softmax(\mathbf{f}_{global}),$$

$$\mathbf{p}_l = softmax(FC(LN(mean(\mathbf{F}_{local})))), \quad (\text{A11})$$

where $LN(\cdot)$ denotes the layer normalization. $FC(\cdot)$ and $\text{softmax}(\cdot)$ represent fully connected layer and softmax operation respectively. Next, we select the label with the maximum score as the final result between global and local. The predicted result p can be formulated as:

$$p = \begin{cases} p_l, & \text{if } \max(\mathbf{p}_g) \leq \max(\mathbf{p}_l), \\ p_g, & \text{if } \max(\mathbf{p}_g) > \max(\mathbf{p}_l), \end{cases} \quad (\text{A12})$$

where $\max(\cdot)$ denotes the maximum score.

Appendix A.5 Training Objective

Stage1: Pre-training modality-specific networks. First, we pre-train the Conv-Transformer3d network of each modality using the cross-entropy loss function. The loss function is defined as:

$$\mathcal{L}_{CE}(\mathbf{p}_g) = -\frac{1}{K} \sum_{i=1}^K \mathbf{y}_i \log(\mathbf{p}_g^i), \quad (\text{A13})$$

where K is the number of subjects. $\mathbf{y}_i$ and $\mathbf{p}_g^i$ represent the ground-truth label and prediction of the modality-specific global feature for the i -th subject, respectively.

Stage2: Fine-tuning to adapt the multi-modal task. We freeze the modality-specific network and only fine-tune CVPL and RTPL modules in the multi-modal branch. The training loss is calculated as follows:

$$\mathcal{L}^{mu} = \frac{1}{n} \sum_{i=1}^n (\mathcal{L}_i^{CSA}) + \mathcal{L}_{CE}(\mathbf{p}_l^{mu}) + \mathcal{L}_{CE}(\mathbf{p}_g^{mu}) + \mathcal{L}_{align}(\mathbf{C}_n^{mu}), \quad (\text{A14})$$

where $\mathbf{p}_g^{mu}$ and $\mathbf{p}_l^{mu}$ denote the prediction results of global and local features in the multi-modal branch. n is the number of transformer blocks in the Conv-Transformer3d network. $\mathbf{C}_n^{mu}$ denotes the output of CVPL. $\mathcal{L}_{CE}$ is defined by Eq. A13. $\mathcal{L}^{CSA}$ is defined by Eq. A3, and $\mathcal{L}_{align}$ is defined by Eq. A10.

Stage3: Local feature representation for the pre-trained unimodal network. To enhance the modality-specific features representation. We add RTPL to the PET and MRI branches. Only the RTPL module can be trained in this stage. The third-stage loss function is represented as:

$$\mathcal{L}^{pet} = \mathcal{L}_{CE}(\mathbf{p}_l^{pet}) + \mathcal{L}_{align}(g_n(\mathbf{H}^{pet})), \quad (\text{A15})$$

$$\mathcal{L}^{mri} = \mathcal{L}_{CE}(\mathbf{p}_l^{mri}) + \mathcal{L}_{align}(g_n(\mathbf{H}^{mri})), \quad (\text{A16})$$

where $\mathbf{p}_l^{pet}$ and $\mathbf{p}_l^{mri}$ represent the classification results of local features in the PET and MRI branches respectively. $g_n(\mathbf{H}^{pet})$ and $g_n(\mathbf{H}^{mri})$ denote the visual feature output from the n -th transform block in the PET and MRI branch separately. The loss function $\mathcal{L}_{align}$ is defined by Eq. A10.

Appendix B Experiments

Appendix B.1 Dataset and Evaluation Metric

Datasets: To evaluate the effectiveness of the proposed method, we conduct experiments on the multi-modal brain disease datasets, including ADNI [9] and ABIDE [10]. The demographic information of the ADNI and ABIDE datasets are shown in Table B1. There are three classes on the ADNI dataset, which are normal controls (NC), mild cognitive impairment (MCI), and Alzheimer's disease (AD). We collect data from 455 patients who have paired 18F-FDG PET and T1 weighted MRI images, including 168 AD, 162 MCI, and 125 NC subjects. For each subject, we select 50 paired images from the middle slices. There are 23.2k paired images in total. Following the works [11, 12], we adopt the AC-PC alignment, registration, normalization, and spatial smoothing operations to preprocess the dataset. Then, we can obtain $50 \times 193 \times 193$ images for each subject. The ABIDE is an autism brain imaging dataset categorized into two groups, including normal controls (NC) and autism spectrum disorder (ASD). We collected data from 895 patients who have paired fMRI and T1 weighted MRI images, including 482 NC and 413 ASD subjects. The ABIDE pre-processing follows the previous studies [13, 14]. We randomly split the data into the training/validation/test sets at a ratio of 80/10/10%. The subjects in the training set are not present in the validation and testing sets. For the missing modality dataset, we randomly select 50% of the data with complete modalities. Then, we randomly select 25% of the data that only includes PET/fMRI modality data and the remaining 25% of the data that only includes MRI modality data. Following previous works [15, 16], we use four commonly used metrics for evaluation: Accuracy (ACC), Sensitivity (SEN), Specificity (SPE), and Area Under the Receiver Operating Characteristic Curve (AUC).

Appendix B.2 Implementation Details

The training process of the proposed method is divided into three stages. In the first stage, we pre-train the Conv-Transformer3d backbone network using unimodal data from the collected dataset. The input size of Conv-Transformer3d is $50 \times 224 \times 224$, where 50 is the frames of each subject, and 224 is the width and height of the input. In the second stage, we fine-tune the CVPL module to obtain global feature representation and the RTPL module to obtain semantic-aware local feature representation from the multi-modal data. In the RTPL module, we hand-crafted expert knowledge prompts

Table B1 Demographic information of the ADNI and ABIDE datasets.

Datasets	label	subjects	images	Age
ADNI	AD	168	8400	74.43 $\pm$ 8.48
	NC	125	6250	76.58 $\pm$ 8.31
	MCI	162	8100	74.15 $\pm$ 86.93
ABIDE	ASD	413	20650	16 $\pm$ 7.8
	NC	482	24100	17.11 $\pm$ 7.04

and encoded them with the ResNet50 CLIP model as $T \in \mathbb{R}^{4 \times 1024}$. In the third stage, we only fine-tune the RTPL module within the unimodal branch to enhance the feature representation. For all stages, the batch size is set to 16, and the training epoch is set to 20. We adopt the AdamW optimizer with an initial learning rate of 1e-4 for training. The experiments are trained on a single NVIDIA RTX3090 GPU. Moreover, we maintain consistent experimental conditions for all comparison methods.

Appendix B.3 Quantitative Comparison

To evaluate the effectiveness of the proposed method, we conduct experiments on ADNI and ABIDE datasets. For the ADNI dataset, we conduct two kinds of classification tasks, including 1) AD & MCI & NC, and 2) AD & NC. For the ABIDE, we perform a classification task for ASD & NC.

Appendix B.3.1 Efficiency and effectiveness of the Conv-Transformer3d network.

In this part, we focus on evaluating the performance of the proposed Conv-Transformer3d backbone network. The backbone network aims to learn the modality-specific features. Therefore, we conduct two classification tasks using unimodal data to evaluate its effectiveness on the ADNI dataset. We compare five backbone networks: ResNet50 [17], vit3d [18], simple-vit [19], vivit [20] and PT-DCN-backbone [12]. The comparative results are shown in Table B2. On the two classification tasks, the proposed Conv-Transformer3d network outperforms other methods, validating the effectiveness of extracting modality-specific features. Moreover, we evaluate the efficiency of the Conv-Transformer3d network. Our Conv-Transformer3d network requires 5.36G floating point operations (Flops) and 14.26M parameters (Params). The results indicate that the Conv-Transformer3d network requires fewer parameters and lower memory usage compared to other backbone networks.

Table B2 The effectiveness and efficiency of the proposed Conv-Transformer3d network.

Methods	Modality	AD&MCI&NC				AD&NC				Flops(G)	Params(M)
		ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE		
ResNet50 [17]	PET	56.00	68.28	58.20	78.34	84.38	84.52	83.33	85.71	6.31	30.73
	MRI	62.00	71.74	71.95	75.57	71.88	72.62	66.67	78.57		
vit3d [18]	PET	48.00	68.18	44.95	72.96	75.00	75.00	66.67	85.71	196.46	39.38
	MRI	56.00	75.57	51.86	77.08	65.62	69.84	88.89	35.71		
simple-vit [19]	PET	52.00	69.74	49.20	75.11	84.38	91.67	100	64.28	196.44	39.37
	MRI	56.00	79.47	51.85	76.04	71.88	75.40	94.44	42.86		
vivit [20]	PET	36.00	63.02	33.33	66.66	56.25	84.13	0	100	195.05	77.18
	MRI	36.00	68.69	33.33	66.66	56.25	61.51	100	0		
PT-DCN-backbone [12]	PET	36.00	58.00	33.33	66.66	71.87	70.23	83.33	57.14	39.22	169.35
	MRI	58.00	74.75	55.28	78.12	56.25	51.58	88.89	14.29		
Conv-Transformer3d	PET	62.00	79.32	60.96	80.90	87.50	95.24	88.89	85.71	5.36	14.26
	MRI	60.00	77.78	60.84	79.28	78.13	83.73	77.78	78.57		

Appendix B.3.2 Comparison with SOTA algorithms.

To evaluate the effectiveness of the proposed framework, we compare our method with other state-of-the-art (SOTA) multi-modal methods under both complete and missing modality settings. In the missing modality setting, the missing modality data is replaced with available modality data for testing. For a fair comparison, we use the collected multi-modal dataset for training in all experiments. The comparison results are obtained using the official code released on GitHub. For the comparison methods without released code, we reproduce them based on their papers. MMGL [16] requires input in the form of feature vectors. In our experiment, we use the pre-trained ResNet50 network to extract the features from different modalities. BLADNet-Cat [21] method concatenates multi-modal features that are extracted from BLADNet networks for disease diagnosis.

Results on ADNI dataset. From Table B3, we can observe that our method achieves the highest performance in both AD&MCI&NC and AD&NC on the complete modality setting. Especially in the AD&MCI&NC task, the proposed method achieves an advantage of over 6%, 0.62%, 5.56%, and 2.78% in ACC, AUC, SEN and SPE, compared to the second-best method. It demonstrates that the proposed method can extract distinctive features, thereby enhancing the performance in the challenging task of AD&MCI&NC. In the missing modality setting, our method achieves the highest performance. Compared to the results with the complete modality setting, the accuracy (ACC) decreases by 6% in the AD&MCI&NC. It validates that the proposed method can effectively handle the issue of missing modalities.

Table B3 Results of mVTPrompt compared with SOTA multi-modal methods on ADNI dataset.

Methods	Modality	Complete modality								Missing modality			
		AD&MCI&NC				AD&NC				AD&MCI&NC			
		ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE
MiSePyNet (2020) [11]	PET	60.00	77.20	58.93	79.39	76.25	75.16	83.89	66.43	-	-	-	-
BLADNet (2023) [21]	PET	62.00	72.12	63.23	81.02	84.36	85.32	77.78	92.86	-	-	-	-
PT-DCN (2021) [12]	PET&MRI	76.00	89.50	75.66	87.84	87.50	87.30	88.89	85.71	56.00	76.47	56.60	78.13
MMGL (2022) [16]	PET&MRI	70.00	78.32	71.67	84.95	81.25	82.53	72.22	92.86	56.00	67.84	57.67	78.01
Conv-eRVFL (2022) [22]	PET&MRI	66.00	77.36	70.55	84.15	81.87	81.98	88.89	82.86	56.00	67.25	56.85	77.66
LLMF (2023) [23]	PET&MRI	76.00	82.82	77.23	88.21	87.50	88.10	83.33	92.86	58.00	69.74	59.58	79.89
BLADNet-Cat (2023) [21]	PET&MRI	70.00	78.04	71.27	84.81	81.25	81.75	77.78	85.71	48.00	61.44	48.33	84.80
MDL-Net (2024) [4]	PET&MRI	68.00	75.70	67.72	83.68	81.25	81.75	77.78	85.71	64.00	78.89	63.49	81.82
HAMMF (2024) [24]	PET&MRI	78.00	89.80	78.56	89.00	87.50	95.23	83.33	92.86	72.00	86.83	71.42	85.74
mVTPrompt	PET&MRI	84.00	90.42	84.12	91.78	90.63	97.22	88.89	92.86	78.00	87.47	78.04	88.77

Results on ABIDE dataset. As shown in Table B4, we report the experimental results on the ABIDE dataset. It can be observed that our method outperforms other SOTA methods, achieving 82.22% in ACC, 84.1% in AUC, 80% in SEN, and 84.44% in SPE. In the missing modality setting, we achieve 75.55% in ACC, 74.81% in AUC, 66.67% in SEN, and 84.44% in SPE. Compared to the results with the complete modality setting, the accuracy (ACC) decreases by 6.67%. These results demonstrate that our method can effectively support input from any modality. Furthermore, we evaluate the efficiency of Our mVTPrompt in comparison with SOTA methods. Among the compared approaches, MMGL, Conv-eRVFL, BLADNet-Cat, and LLMF are 2D-based methods, while the remaining methods are 3D-based. Since 2D methods operate on individual image slices, we evaluate their computational cost based on a single slice. Our mVTPrompt requires 12.76G floating point operations (Flops) and 39.06M number of parameters (Params).

Table B4 Results of mVTPrompt with SOTA multi-modal methods on ABIDE dataset.

Methods	Modality	Complete modality				Missing modality				Flops(G)	Params(M)
		ASD&NC				ASD&NC					
		ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE		
MMGL (2022) [16]	fMRI&MRI	81.11	81.11	75.55	86.66	72.22	72.22	75.55	68.88	25.23	61.58
Conv-eRVFL (2022) [22]	fMRI&MRI	61.11	61.11	31.11	91.11	51.11	51.11	13.33	88.88	0.06	0.18
BLADNet-Cat (2023) [21]	fMRI&MRI	81.11	81.11	77.78	84.44	66.67	66.67	68.88	64.44	1.41	3.24
LLMF (2023) [23]	fMRI&MRI	67.78	67.78	64.44	71.11	58.89	58.89	46.67	71.11	17.36	100.3
MDL-Net (2024) [4]	fMRI&MRI	67.80	71.65	66.67	68.89	50.00	54.44	100	0	6.28	2.64
HAMMF (2024) [24]	fMRI&MRI	71.11	67.21	46.67	95.55	64.44	66.76	64.44	64.44	19.29	17.16
mVTPrompt	fMRI&MRI	82.22	84.10	80.00	84.44	75.55	74.81	66.67	84.44	12.76	39.06

Appendix B.4 Discussion

Appendix B.4.1 Effectiveness of different components

To verify the effectiveness of different components of the proposed method, we perform ablation experiments on the ADNI and ABIDE datasets as demonstrated in Table B5. “Global” indicates the global feature. “Local” indicates the local feature, which is output from the RTPL module. The selection of “Global and Local” refers to the results obtained through a decision-level fusion method. There are three branches of the proposed method that can cope with input from any modality. In the first group, we show the ablation studies from the PET/fMRI branch, denoted as rows 1, 2, and 3. In the second group, we demonstrate the results from the branch of MRI, denoted as rows 4, 5, and 6. In the third group, we utilize paired multi-modal data and report the results from the multi-modal branch displayed in rows 7, 8, and 9.

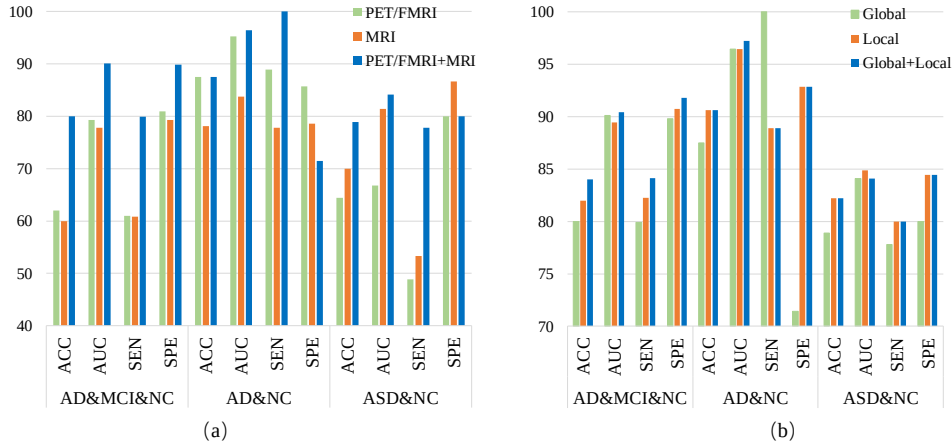
Analysis of CVPL module. For rows 1 and 4, we only use the Conv-Transformer3d network to obtain the classification results from unimodal. After introducing the Cross-modality Visual Prompt Learning module (CVPL) as row 9, we achieve an ACC of 84% in the AD&MCI&NC task, which is 18% higher than only using PET data and 20% higher than only using MRI data. The comparison results as shown in Figure B5 (a) demonstrate that the CVPL module provides a general

solution to fine-tune pre-trained unimodal models for adapting the multi-modal tasks, fully exploring the complementary information across different modalities.

Analysis of RTPL module. Rows 1, 4, and 7 denote the global features output from different branches. Rows 2, 5, and 8 represent the local features output from the RTPL module. Rows 3, 6, and 9 present the results of integrating local and global features at the decision level. Compared to the “Global”, the “Local” achieves consistent improvement across all classification tasks, which demonstrates that paying more attention to the distinctive local features is necessary to enhance classification performance. By integrating both local and global features at the decision level, we observe further performance improvements: ACC increases by 2%, AUC by 0.99%, SEN by 1.85% and SPE by 1.04% in the AD&MCI&NC classification task of the multi-modal branch. These results indicate that both local and global features are crucial for more reliable brain disease diagnosis.

Table B5 Ablation studies of the proposed method on the ADNI and ABIDE datasets. Figure (a) represents the performance of the global features from different branches. Figure (b) shows the ablation results from the multi-modal branch.

Methods				ADNI								ABIDE			
				AD&MCI&NC				AD&NC				ASD&NC			
PET	MRI	Global	Local	ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE
✓		✓		62.00	79.32	60.96	80.90	87.50	95.24	88.89	85.71	64.44	66.81	48.88	80.00
✓			✓	64.00	81.91	61.90	81.48	90.62	96.03	100	78.57	66.67	68.49	42.22	91.11
✓		✓	✓	64.00	80.60	62.96	81.60	90.62	96.03	100	78.57	66.67	64.59	42.22	91.11
	✓	✓		60.00	77.78	60.84	79.28	78.13	83.73	77.78	78.57	70.00	81.43	53.33	86.67
		✓	✓	64.00	79.48	62.96	81.71	78.13	84.52	66.67	92.85	75.55	80.98	73.33	77.78
	✓	✓	✓	66.00	77.19	66.40	83.10	81.25	80.55	77.78	85.71	76.67	81.67	73.33	80.00
✓	✓	✓		80.00	90.11	79.90	89.81	87.50	96.43	100	71.43	78.90	84.10	77.78	80.00
✓	✓		✓	82.00	89.43	82.27	90.74	90.63	96.43	88.89	92.86	82.22	84.89	80.00	84.44
✓	✓	✓	✓	84.00	90.42	84.12	91.78	90.63	97.22	88.89	92.86	82.22	84.10	80.00	84.44



Appendix B.4.2 Visualization

Visualization of the classification results. We use 2D t-SNE [25] to visualize the distribution of classifiers from different branches, as shown in Figure B1. In the first row, we show the results of the AD&MCI&NC task on ADNI dataset. It is difficult for the PET classifier to distinguish between NC and MCI classes, while the MRI classifier also faces challenges in distinguishing the NC and AD classes. However, after introducing the Cross-modality Visual Prompt Learning module (CVPL), we are able to distinguish the NC, MCI, and AD categories more effectively. In the second row, we present the classification results on ABIDE dataset. For the classifier of fMRI, we observe difficulty in distinguishing between the ASD and NC classes. The branch of MRI can distinguish the difference between the classes, but the inter-class distance remains small. After introducing the CVPL module, we achieve a larger inter-class distance. These results demonstrate that the proposed CVPL module successfully captures complementary information from different modalities, thereby improving the performance of disease diagnosis.

Visualization of RTPL. To evaluate that the expert knowledge can guide the network to focus on distinctive regions in RTPL, we use GradCam [26] to visualize the activation maps for the projection matrix $\mathbf{P}^v$ in Eq. A8, as shown in Figure B2. In the visualizations, red indicates higher weights, while blue represents lower weights. In RTPL, we hand-craft four expert knowledge prompts: “Temporal lobe glucose metabolism status; Parietal regions glucose metabolism status; Hippocampus region status; Frontal lobe region status;”. Figure B2 represents the activation maps of $\mathbf{P}_i^v$ corresponding

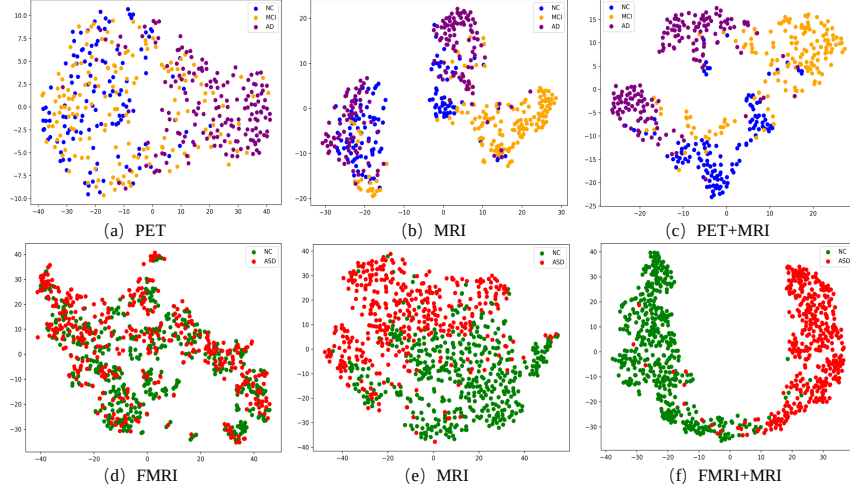


Figure B1 2D t-SNE visualization from different branches on ADNI and ABIDE datasets.

to each expert knowledge prompt. The results demonstrate that the proposed RTPL promotes the network focus on the distinctive regions in the brain, such as the temporal lobe and hippocampus.

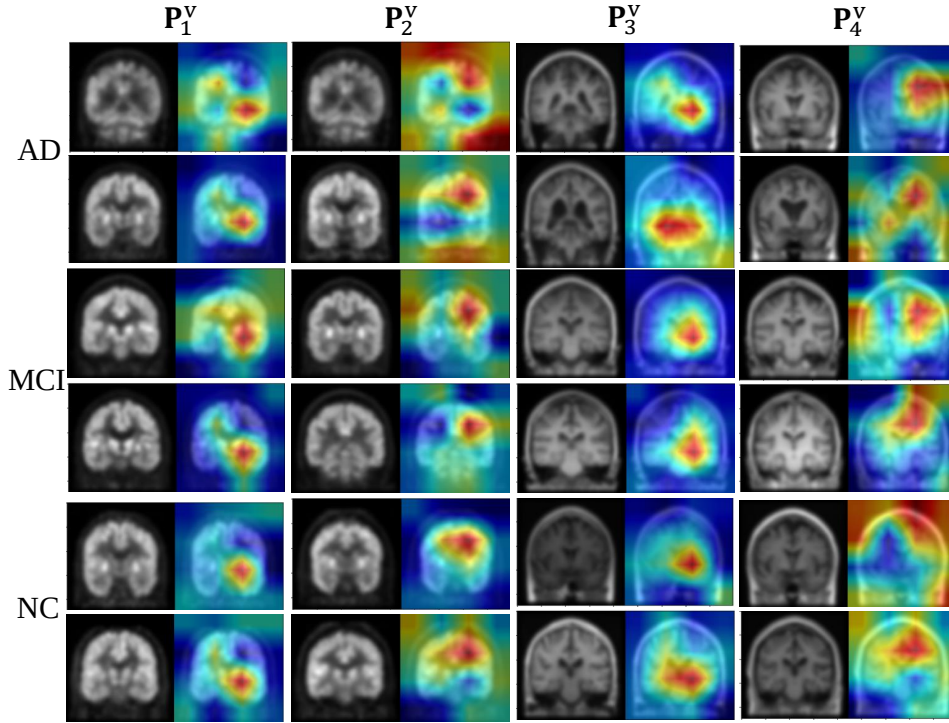


Figure B2 Visualizing the activation maps for the RTPL module from AD, MCI, and NC patients.

Appendix B.4.3 Effects of the number of expert knowledge prompts.

We conduct experiments on the AD&MCI&NC task to evaluate the impact of varying the number of expert knowledge prompts in the RTPL module. First, we add two additional expert knowledge prompts to the original hand-crafted set: “Cingulate Gyrus glucose metabolism status” and “Ventricular region status”, denoted as ‘6’. Then we remove two prompts from the original hand-crafted set: “Parietal regions glucose metabolism status” and “Frontal lobe region status”, referred to as ‘2’. ‘0’ represents the absence of the RTPL module from the multi-modal branch. The accuracy improves when integrated with RTPL, indicating that focusing on distinctive regions is essential for enhancing the diagnosis accuracy. However, adding too many prompts to the RTPL module may force the model to distribute its attention across multiple regions, reducing its focus on key areas and leading to a decrease in AUC.

Table B6 The impact of the number of expert knowledge prompts on the AD&MCI&NC task.

Number	ACC	AUC	SEN	SPE
6	82.00	88.59	82.27	90.97
4	82.00	89.43	82.27	90.74
2	82.00	87.95	82.27	90.97
0	80.00	90.11	79.90	89.81

Appendix C Conclusion

In this paper, we propose a novel training-efficient multi-modal visual and text prompt learning framework for brain disease diagnosis. The proposed method supports the input of complete modality and missing modality data. We design a memory-efficient backbone network named Conv-Transformer3d, which can capture the long-range relationship among different slices. Additionally, we present a novel region and text prompt learning module by introducing expert knowledge into the visual learning network to capture the discriminative features for disease diagnosis. Specifically, we provide a general solution to fine-tune pre-trained unimodal models for adapting the multi-modal tasks, fully exploring the complementary information across different modalities. The effectiveness is validated on both ADNI and ABIDE datasets. Furthermore, we construct a missing modality dataset to evaluate the effectiveness of the proposed method. In our experiments, we adopt a simple method to fuse the global and local information. In the future, we aim to explore more efficient fusion methods for the local and global features, while also investigating a new method to address the challenge of missing modalities during the training and test phases.

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